Ziad Khaled Mahrous Sobh

Technical Office Mechanical Engineer

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OBJECTIVE

Technical Office & BIM Mechanical Engineer with a bachelor's degree in mechanical engineering and over two years of hands-on experience developing coordinated mechanical models, reviewing design packages across SD and DD stages, preparing shop drawings, and extracting quantities for design-and-build and BIM coordination projects. Strong knowledge of HVAC, Firefighting, Plumbing, and Medical Gas systems, with advanced proficiency in Revit MEP and Navisworks.

EDUCATION

Bachelor's degree of mechanical power engineering, Grade: Very Good with honors,

Damietta University May 2024

Graduation Project: Solar powered portable desalination unit by using air humidification, **Grade: Excellent**

EXPERIENCE

Mechanical BIM Engineer – BEC ARABIA LLC (Technical Office) – District 5 May 2025 – Still Now

Roles and Responsibilities:

- Developed 3D mechanical models (HVAC, plumbing, Fire Fighting, Medical Gases) using Revit MEP.
- Reviewing consultant design drawings and preparing detailed Shop Drawings in compliance with project requirements.
- Prepared construction drawings, schedules, and material take-offs.

BIM Engineer – BIM WORKS LLC (Technical Office) – Nasr City Oct 2024 – May 2025

Roles and Responsibilities:

- Developed 3D mechanical models (HVAC, plumbing, Fire Fighting, Machinery) using Revit MEP.
- Conducted clash detection and resolved conflicts using Navisworks.
- Upgrade the Level of Development (LOD) of multiple projects from LOD 200 to LOD 500, ensuring models are detailed and suitable for fabrication and installation.

MEP Engineer – ASEC ENGINEERING CONSULTANTS – Nasr City Feb 2024 – Oct 2024

Roles and Responsibilities:

- HVAC Design: Optimize systems and perform load calculations.
- Firefighting Systems: Design compliant sprinkler systems.
- Plumbing Design: Create water supply and drainage layouts.
- Medical Gas Systems: Design piping for healthcare facilities.
- Collaboration: Coordinate with project teams.

PROJECTS

DARIYA (BEC ARABIA) – Dariya-KSA

- Design and build mixed-use development comprising a **Hotel, and Commercial Buildings**.
- Reviewed and coordinated mechanical (HVAC, Firefighting, Plumbing) design packages across the SD 50%, SD 100%, DD 50%, and DD 100% design stages.
- Tracked design progression against project requirements and flagged coordination issues between MEP disciplines at each design milestone.

BUSTAN PARK (BEC ARABIA) – KSA

- Prepared execution drawings for **HVAC, Fire Fighting, Plumbing, and builder work** across all MEP disciplines.
- Ensured accurate and fully coordinated shop drawings aligned with project requirements.

AMALAA CIVIC ASSETS (BEC ARABIA) – Amalaa, KSA

- Responsible for producing execution drawings and coordinated layouts for buildings and pump rooms.
- Developed and submitted LOD 500 BIM models for mechanical works with high precision.

AMALAA STAFF VALLIGE (BEC ARABIA) – Amalaa, KSA

- Delivered LOD 500 BIM models for multiple residential buildings.
- Prepared accurate mechanical asset data for the project's asset register.
- Responsible for full coordination between all MEP and architectural/structural disciplines.

THARBA MEDICAL CENTER (BEC ARABIA) – Alula, KSA

- Prepared and delivered complete as-built MEP drawings for the project.
- Developed and submitted a highly accurate LOD 500 BIM model for all mechanical systems.

Moral Tower (Outsource) – Dubai, UAE

- Prepared Shop Drawings for the Smoke and Ventilation System as part of remote outsourced work.

Metro Station 3rd Line (BIMWORKS LLS) – Cairo, EGY

- Assisted the MEP team in developing BIM models aligned with point cloud data.
- Modeling and Coordination all Discipline of Mechanical (Firefighting, Plumbing) with other Discipline according to point cloud and Cad drawing.

Adahi Building (BIMWORKS LLS) – Makka, KSA

- Modeling and Coordination all Discipline of Mechanical (HVAC, Firefighting, Plumbing, Machinery) with other Discipline according to point cloud and Cad drawing.
- Upgrade the Level of Development (LOD) of multiple projects from LOD 200 to LOD 500
- Family Creation for most Machinery according to Recap Files.

Bedaya Hospital – Training

- MEP design and BIM modeling using Revit for HVAC, Firefighting, Plumbing, Medical Gases

Villa Moosa – Training

- Prepared HVAC and firefighting schematic design

COURSES

BIM Diploma | Niqat Global Feb 2025 – Nov 2025

Currently enrolled in a 9-month BIM diploma covering LOD100–LOD500 and BIM dimensions (3D – 7D), with hands-on training in Revit, Civil 3D, Dynamo, Navisworks, ACC, and Recap. Attended interactive sessions led by industry experts, focusing on advanced BIM applications and workflow

MEP Diploma | ENG.X Mar 2024 – Oct 2024

A comprehensive Diploma in MEP systems, focusing on practical skills and software proficiency.

Key areas covered include:

- Proficiency in AutoCAD, Revit, Duct Sizer, HAP, and Elite.
- HVAC systems design and installation.
- Firefighting systems design and installation.
- Plumbing systems design and installation.
- The graduation project was to design a hospital from scratch for HVAC, firefighting, plumbing and medical gas systems.

SKILLS

- Revit
- Navisworks
- AutoCAD
- SolidWorks
- HAP
- Microsoft Office
- Duct & Pipe Sizer
- Elite
- Enscape
- Recap

LANGUAGE

- Arabic (Native)
- English (Advanced)